3867. Sum of GCD of Formed Pairs

Solved

Medium Topics Hint

You are given an integer array `nums` of length `n`.

Construct an array `prefixGcd` where for each index `i`:

- Let mx_i = max(nums[0], nums[1], ..., nums[i]).
- prefixGcd[i] = gcd(nums[i], mx_i).

After constructing `prefixGcd`:

- Sort `prefixGcd` in **non-decreasing** order.
- Form pairs by taking the **smallest unpaired** element and the **largest unpaired** element.
- Repeat this process until no more pairs can be formed.
- For each formed pair, **compute** the `gcd` of the two elements.
- If `n` is odd, the **middle** element in the `prefixGcd` array remains **unpaired** and should be ignored.

Return an integer denoting the **sum of the GCD** values of all formed pairs.

The term `gcd(a, b)` denotes the **greatest common divisor** of `a` and `b`.

Example 1:

> **Input:** nums = [2,6,4]
>
> **Output:** 2
>
> **Explanation:**
>
> Construct `prefixGcd`:
>
> | i | nums[i] | mx_i | prefixGcd[i] |
> |---|---|---|---|
> | 0 | 2 | 2 | 2 |
> | 1 | 6 | 6 | 6 |
> | 2 | 4 | 6 | 2 |
>
> `prefixGcd = [2, 6, 2]`. After sorting, it forms `[2, 2, 6]`.
>
> Pair the smallest and largest elements: `gcd(2, 6) = 2`. The remaining middle element 2 is ignored. Thus, the sum is 2.

Example 2:

> **Input:** nums = [3,6,2,8]
>
> **Output:** 5
>
> **Explanation:**
>
> Construct `prefixGcd`:
>
> | i | nums[i] | mx_i | prefixGcd[i] |
> |---|---|---|---|
> | 0 | 3 | 3 | 3 |
> | 1 | 6 | 6 | 6 |
> | 2 | 2 | 6 | 2 |
> | 3 | 8 | 8 | 8 |
>
> `prefixGcd = [3, 6, 2, 8]`. After sorting, it forms `[2, 3, 6, 8]`.
>
> Form pairs: `gcd(2, 8) = 2` and `gcd(3, 6) = 3`. Thus, the sum is `2 + 3 = 5`.

Constraints:

- $1 <= n == nums.length <= 10^5$
- $1 <= nums[i] <= 10^9$

Seen this question in a real interview before? 1/6

Yes No

Accepted 30,651/46.7K | Acceptance Rate 65.7%

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (21)

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